

Digital Research Toolkit for Linguists

Week 3: Looking at data

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Psycholinguistics and Cognitive Modeling Lab

Questions?

Why this “pointless” homework assignment?

- ✓ Successfully install R and RStudio
- ✓ Open and look at RStudio
- ✓ Install and load packages
- ✓ Solved your problems
- ✓ Follow instructions
- ✓ Do what all programmers do with new IDE

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Data types, formats, and encoding

Data types: What the researcher sees

-  response
-  reaction times
-  acceptability judgments
-  completion
-  recordings
-  transcription, annotation
-  brain waves
-  texts
-  video
-  images
-  ...

→ text files (txt, csv, tsv)

Data types: What the statistician sees



nominal

ordinal

interval

ratio



marital status, religion

grades, energy efficiency classes

IQ, temperature in C and F

reaction times, population

Data: What R sees

logical	TRUE
integer	1 or 1L
double	1.0
complex	1+0i
character	"one"
double "not a number"	NaN
double "infinity"	Inf
logical "missing" value	NA
special variable without a type	NULL
...	

How does our data look like?

What question does this experiment try to answer?

Right or wrong answer:

Question ID:

Response time:

Answer:

Logical

Integer

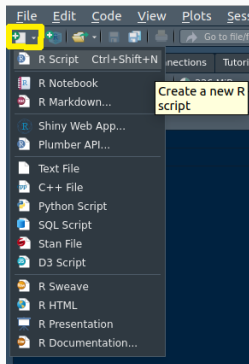
Double

Character

Unsure? `typeof(1L)` or `is.numeric(1)`

Certain? `as.character(1)`

Let's get comfortable with R first. Open your
• R script from last week.



Hello world

```
print("Hello World")  
print("Twas brillig,  
and the slithy toves  
did gyre and gimble  
in the wabe: all mimsy  
were the borogoves,  
and the mome raths  
outgrabe.")
```



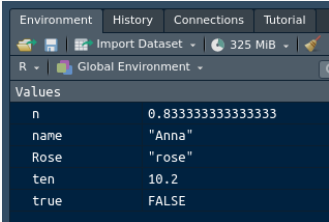
Assignment

Saves information for later, keeps it in the environment, but **overwrites existing values**. These operators differ in scope.

```
x <- value          ten <- 10.2
value -> x          "rose" -> Rose

x = value           name = "Anna"

x <<- value         true <<- FALSE
value ->> x          13/12 ->> n
```



The screenshot shows the R Studio Environment pane with the following content:

Values	
n	0.833333333333333
name	"Anna"
Rose	"rose"
ten	10.2
true	FALSE

```
jabberwocky <- print("Twas brillig ...")
```

❓ What are some good names?

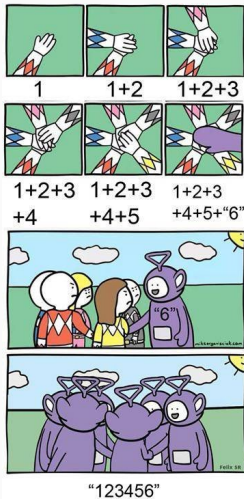
Coercion or implicit type conversion

When R combines different data types, it coerces simpler types to more complex types. The conversion follows a hierarchy:

Logical → Integer → Double → Character

<code>TRUE + 1 = 2</code>	integer
<code>5L + 2 = 7</code>	integer
<code>3.7 * 3L = 11.1</code>	double
<code>99999.0e-1 - 3.3e+3 = 9999.9 - 3300 = 6699.9</code>	double
<code>10 / as.complex(2) = 5+0i</code>	complex
<code>as.character(5) / 5 = </code>	non-numeric argument!
<code>paste(5+0i, "five") = "5+0i five"</code>	character

Lost in translation: How to mess up data



Make different types of entries in the same column.

惛憊檔微豎拔淡瘡湯堆酒灯濛浮獵步燐峇派
 斂穀枚習啖妨浹娠敷果H編 咳。槍奏徽棚愁物
 僅映 整豎<散率時撓漣·狹蠱 寐·茵嫗·
 茵β茵嫗·銑茵嫗·茵嫗·茵嫗·茵嫗·茵嫗·
 鯉茵嫗·茵嫗·茵嫗·茵嫗·茵嫗·茵嫗·茵嫗·
 鯉茵嫗·茵嫗·茵嫗·茵嫗·茵嫗·茵嫗·茵嫗·

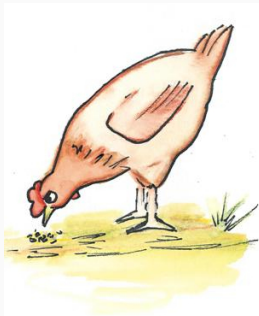
Falsches Äben von Xylophonmusik
 quÄ=lt jeden grÄ¶ÄYeren Zwerg.
 Dis aux filles de faire la fÄete
 Ä<0xa0> l'heure du cinq Ä<0xa0>
 sept.
 äf'äf<0x90>äf<0xa0>äf-äffäffäfi
 äf'äf<0x90>äf<0xa0>äf"äf<0x90>
 äfiäfäfäfäf<0x90>
 äf"äfäfäf<0x90>äf-äf" äffäff<0x90>äf
 zäff<0x90>äf<0xa0>äf<0x90>äf"äf
 <0x9d>äf'äf-?

Falsches üben von Xylophonmusik
 quv\$lt jeden grv÷üeren Zwerg.
 Dis aux filles de faire la fÿnte
 vÿ l'heure du cinq vÿ sept.
 SE·SEZSEtSEöSEfSEöSE"
 SEiSEZSEtSEdSEZ SE·SEöSESEZ
 SEöSEüSEZSEöSEE
 SEöSEZSEöSEZSEtSEZSEöSEöSEöSEö?

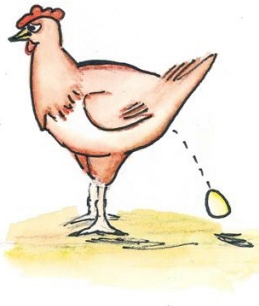
Falsches üben von Xylophonmusik
 quält jeden größeren Zwerg.
 Dis aux filles de faire la fête à
 l'heure du cinq à sept.
 ქართუღს გარდა სხვა ენაზე
 რაპარაკობ?

Make or change the encoding to something random.

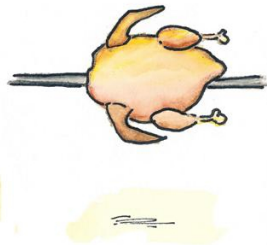
Lost in translation: How NOT to mess up data



Input



Output



Kaputt

- Be careful and know what you put in
- Be mindful of the character encoding (when in doubt, UTF-8)

Inspecting data

Look at what you did



U.s.a



U.k



China

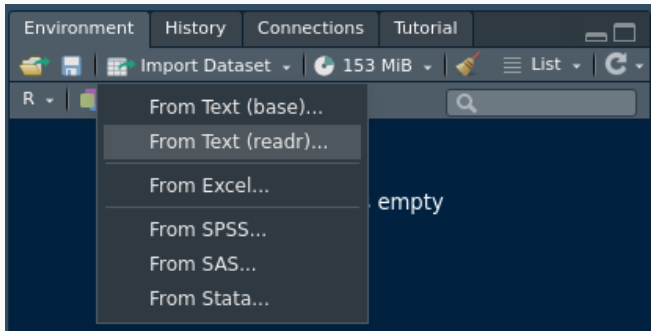


Stuttgart

Reading in data

- Download `moses_raw_data.csv` from ILIAS. Save it to the working directory.
- Open your R script & load the packages: `tidyverse`, `psych`
- Read in the file `moses_raw_data.csv` and assign it to the name `moses`.

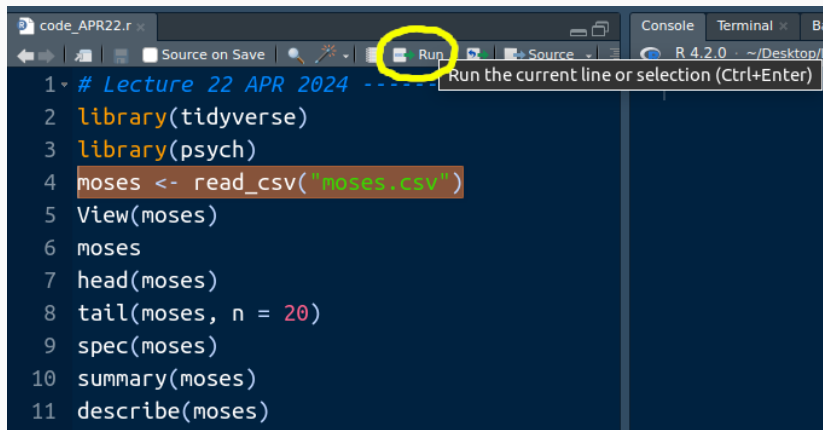
Select Environment > Import Dataset > readr



Load the data into environment

```
moses <- read_csv("moses_raw_data.csv")
```

CTRL+ENTER or CMD+RETURN or click on “Run”



```
code_APR22.r x
Source on Save
Run
Run the current line or selection (Ctrl+Enter)
1 # Lecture 22 APR 2024 -----
2 library(tidyverse)
3 library(psych)
4 moses <- read_csv("moses.csv")
5 View(moses)
6 moses
7 head(moses)
8 tail(moses, n = 20)
9 spec(moses)
10 summary(moses)
11 describe(moses)
```

You now have a **data frame** or **tibble** called **moses**.

Must know shortcuts for lazy programmers

	Windows/Linux	Mac
use auto-completion	TAB	TAB
run selected code	CTRL+ENTER	CMD+RETURN
leave comments #	CTRL+SHIFT+C	CMD+SHIFT+C
console back & forth	↑↓	↑↓
make sections	CTRL+SHIFT+R	CMD+SHIFT+R
mane new script	CTRL+SHIFT+N	CMD+SHIFT+N

Look at what you did

<code>View(moses)</code>	in the RStudio window
<code>moses</code>	in the console
<code>print(moses, n=Inf)</code>	in the console
<code>head(moses)</code>	first 6 rows
<code>tail(moses, n=20)</code>	last 20 rows
<code>spec(moses)</code>	column properties
<code>summary(moses)</code>	summary statistics
<code>describe(moses)</code>	summary statistics vol. 2
<code>colnames(moses)</code>	column names
<code>summary(NAME)</code>	→ calling function with one argument
<code>head(NAME, n=20)</code>	→ calling function with two arguments
<code>dbinom(x=6, size=9, prob=0.5)</code>	3 named arguments in order

A meme featuring Ned Stark from the TV show Game of Thrones. He is shown from the chest up, wearing his iconic black fur cloak with a thick fur collar. He has a beard and is looking slightly to his left with a serious expression. He is holding a silver microphone in his right hand, as if he is about to give a speech or announcement. The background is a plain, light grey.

BRACE YOURSELF

STATS ARE COMING

DIYLOL.COM

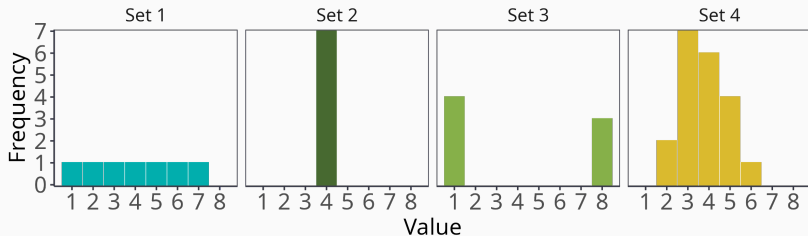
Summarize

Min.	min()	minimal value
Max.	max()	maximal value
Mean	mean()	average
1st Qu.	quantile()	25%
Median	quantile()	middle number == 2nd quantile == 50%
3rd Qu.	quantile()	75%
NA's	TBA	nr missing data

Describe

	<code>colnames()</code>	item name
<code>vars</code>	<code>colnames()</code>	item number
<code>n</code>	<code>n()</code>	number of valid cases
<code>mean</code>	<code>mean()</code>	mean
<code>median</code>	<code>median()</code>	median
<code>min</code>	<code>min()</code>	minimum
<code>max</code>	<code>max()</code>	maximum
<code>range</code>	<code>range()</code>	range
<code>sd</code>	<code>sd()</code>	standard deviation ($\sqrt{\text{variance}}$)
<code>trimmed</code>		trimmed mean
<code>mad</code>		median absolute deviation
<code>skew</code>	<code>skew()</code>	skew
<code>kurtosis</code>	<code>kurtosis()</code>	kurtosis
<code>se</code>	<code>mean_se()</code>	standard error

Central tendency



	Set 1	Set 2	Set 3	Set 4
Values	1, 2, 3, 4, 5, 6, 7	4, 4, 4, 4, 4, 4, 4	1, 1, 1, 1, 8, 8, 8	2, 2, 3, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 5, 5, 5, 5, 6
N	7	7	7	20
Mean	4	4	4	4
Median	4	4	1	4
SD	2	0	4	1

What a information mess!

Too much

native, Instructions

Too little

condition 1? 2?? 100???

Missing

NA

Inconsistent:

❓ Margaret Thatcher was the former **president/prime minister** of which country?

💬 Don't know, Great Britain, UK, United Kingdom, can't answer, england, the UK, uk, Prime Minister of UK, can't know, ...

❓ According to the Bible, how many animals of each kind did **Moses/Noah** take on the ark?

💬 2, Don't know, Two, can't answer, don't know, don't know, no idea, two, 2 of each kind, 2%2C 1 male 1 female, 42, 62, ...



Clean up after yourselves

select meaningful columns	<code>select(WHERE, WHAT)</code>
remove missing values	<code>na.omit(WHERE)</code>
choose or remove data	<code>filter(WHERE, TRUE CONDITION)</code>
reorder values	<code>arrange(WHERE, HOW)</code>
rename columns	<code>rename(WHERE, NEW = OLD)</code>
create values	<code>mutate(WHERE, NEW = FUNCTION(OLD))</code>

`=` `!=` `==`

`=` is assignment, `==` is equality

Functions are executed, results are displayed, but `nothing is saved`.

Wrap-up

Summary

- ✓ scripts
- ✓ data types
- ✓ encoding
- ✓ reading in data
- ✓ inspecting data
- ▶▶ Hands-on data cleanup, basic R operations, tidy code, data manipulation.

Homework assignment due April 28th 12:00

- ❓ Read chapters 3, 4 and 5 of *R for Data Science* (Wickham et al. 2023)
- ❓ Complete assignment 3 (→ ILIAS)
- ❗ Name your files with name and assignment number, e.g. `pryslopskaassignment3.R`

References

-  Reisinger, Emil C (June 2009). ***“Bloggewriter” – Gastbeitrag Prof. Dr. Reisinger.*** Guest post introduced by Helmut Wicht. URL: <https://scilogs.spektrum.de/anatomisches-allerlei/bloggewriter-gastbeitrag-prof-dr-reisinger/>.
-  Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund (2023). ***R for data science: import, tidy, transform, visualize, and model data.*** 2nd ed. O'Reilly Media, Inc. URL: <https://r4ds.hadley.nz/>.